

ROAM: GLOSSARY

MATH FACTS



Addition		
Problem Type	Description	Example
Less than 10	Sum of two single-digit numbers with a solution less than 10.	$3 + 2$
Doubles	Sum of two equal numbers.	$3 + 3$
10 to 20	Sum of two numbers with a solution between 10 and 20.	$7 + 5$
To 10	Sum of two single-digit numbers with the solution being 10.	$4 + 6$
Subtraction		
Problem Type	Description	Example
One smaller	Difference of two numbers with the solution being 1.	$5 - 4$
Doubles	Difference of two numbers with the solution being half of the minuend.	$6 - 3$
Multiplication & Division		
Problem Type	Description	Examples
1s facts 2s facts : : 10s facts	Products and quotients within the 1s-10s fact families .	2×5 Problem Type: 2s facts, 5s facts*
		$35 \div 7$ Problem Type: 7s facts*
Perfect squares	Perfect square operation with two numbers 1-10, showing either: - A product of two equal factors - A quotient equal to the divisor	6×6 $36 \div 6$

*The subscore table categorizes multiplication problems by both factors and division problems by the divisor based on common practice within math curricula. In the subscore table, Problem Types to Work On reports only the top 3 assessed math facts across multiplication and division that require additional support.



Addition, Subtraction, Multiplication, and Division		
Problem Type*	Description	Examples
Without regrouping	Calculations contained entirely within their respective place values .	$\begin{array}{r} 21 \\ + 12 \\ \hline \end{array}$
		$\begin{array}{r} 35 \\ - 13 \\ \hline \end{array}$
With regrouping	Calculations requiring a transfer across place values.	$\begin{array}{r} 12 \\ \times 2 \\ \hline \end{array}$
		$\begin{array}{r} 44 \\ \div 4 \\ \hline \end{array}$
<i>*Note: Bolded keywords and other important terms can be found in the ROAM Keywords and Terms Glossary. </i>		

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KEY WORDS AND TERMS



Addend

Any of the numbers being totalled in an addition problem (e.g., both 3 and 7 in $3 + 7 = 10$).

Algorithm

A step-by-step procedure or set of rules designed to solve a specific problem or perform a computation.

Difference

The result of subtraction.

Dividend

The number that is being divided in a division problem (e.g., the 25 in $25 \div 5$).

Fact Family

A group of related equations that all use the same numbers (e.g., $1 + 3 = 4$; $3 + 1 = 4$; $4 - 1 = 3$; and $4 - 3 = 1$).

Math Fluency

Math fluency is the ability to efficiently, accurately, and relatively effortlessly produce the solutions for single-digit math problems through mental processes such as memory/retrieval and calculations.

Partial Product

Each product formed in the intermediate steps of completing a multiplication problem (e.g. 32×4 has the partial products of 120 and 8).

Partial Quotient

Each quotient formed in the intermediate steps of completing a division problem (e.g. $63 \div 3$ has the partial quotients of 20 and 1).

Perfect Square

A product involving the multiplication of two equal numbers (e.g., $7 \times 7 = 49$; $5 \times 5 = 25$).

Place Value

The value of a digit of numbers. For example, ones, tens, hundreds in the decimal counting

Product

A quantity obtained by multiplication of two or more numbers.

Quotient

The number resulting from the division of one number by another.

Rank Order

To arrange scores in sequential order to identify similar skill levels and create instructional groups.

Regrouping:

To rearrange numbers between place values (ones, tens, hundreds, etc.) while keeping the total value the same. This is sometimes referred to as “carrying” in addition and “borrowing” in subtraction. ROAM uses “regrouping” in place of carrying/borrowing in alignment with [best practice research](#) for math education and [English Language Learners](#).

- **Addition & Subtraction:** Regrouping occurs when there is an exchange of place value as numbers are composed or decomposed.
- **Multiplication & Division:** In multiplication, regrouping occurs when at least one partial product results in a two digit number, requiring you to regroup to the next place value. In division, regrouping occurs when at least one partial quotient results in a remainder, requiring you to regroup to the next place value. For examples, see [ROAM Glossary: Calculation Fluency](#).

Retrieval

The process of determining a solution to an arithmetic problem from memory (often rapidly), without having to apply a computation-based strategy.

Solution

A general term used for the correct response or answer to a math problem (includes: sum, difference, product, and quotient).

Sum

A quantity obtained by addition.